Comparison of intraoperative functional brain maps obtained with RGB imaging with preoperative functional Magnetic Resonance Imaging maps: a task-based and resting-state study

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ABSTRACT

RGB optical imaging is a marker-free, contactless, and non-invasive technique that is able to monitor hemodynamic brain response following neuronal activation using task-based and resting-state procedures. As opposed to functional task-based analyses, resting-state functional connectivity aims to identify the low frequency cortical hemodynamic fluctuations during patient rest that are linked to resting-state networks. Using intraoperative optical imaging, the main issues of using resting-state procedures come from the partial access to the brain cortex, whereas fMRI or fNIRS resting-state models used whole brain imaging. Task-based fMRI brain maps were compared to intraoperative optical functional brain maps by registering these maps to a preoperative anatomical MRI volume. The objective is to improve the patient care process before, during and after neurosurgery. With the task-based procedure, the RGB brain map showed a good correspondence with task-based fMRI ($DICE = 0.75$). With the resting-state procedure, the RGB brain map showed a good correspondence with task-based fMRI (seed correlation method: $DICE = 0.58$ and ICA method: $DICE = 0.75$).

Keywords: Intraoperative imaging; Optical imaging; RGB imaging ; Preoperative imaging; fMRI

1. INTRODUCTION

Non-invasive functional brain mapping is an imaging technique that allows the location of functional areas of the patient's brain. This technique is used during brain tumor resection surgery to indicate to the neurosurgeon the cortical tissues which should not be removed without cognitive impairment. Functional magnetic resonance imaging (fMRI)[1] is the preoperative gold standard for identifying the patient's functional areas. However, after the patient's craniotomy, a brain shift invalidates the relevance of neuronavigation to localise functional areas during surgery.[2] To avoid localisation errors, intraoperative MRI has been suggested, but it complicates the surgical procedure and is therefore rarely used. For these reasons, Electrical brain stimulation (EBS)[3] is preferred during neurosurgery, but this technique is mainly limited by its low spatial resolution ($\approx 5mm$[4]) and has the potential risk to trigger epileptic seizures. Optical imaging provides an ideal solution for intraoperative functional brain mapping because the analysis of the light absorption allows to monitor the brain activity (motor or sensory tasks for example) with quantification of the concentration changes in oxy- (ΔC_{HbO_2}) and deoxygenated hemoglobin (ΔC_{Hb}) in brain cortex.[5–9]

In neuroimaging applications, functional brain areas can be identified using task-based or resting-state procedures. As opposed to functional task-based analyses, resting-state functional connectivity aims to identify the

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low frequency cortical hemodynamic fluctuations (< 0.1Hz) that reflect the patient neuronal activity at rest and that are linked to resting-state networks.[10] Using intraoperative optical imaging, the main issues of using resting-state procedures come from the partial access to the brain cortex, whereas fMRI or fNIRS resting-state models used whole brain imaging.

In this study, we propose to compare preoperative functional brain maps obtained with fMRI to intraoperative functional brain maps obtained with RGB imaging. For this purpose a multi-modality registration procedure was set up. Functional MRI and optical maps were co-registered to a preoperative anatomical MRI volume. This registration procedure may be a great tool to improve the patient care process before, during and after neurosurgery.

2. MATERIAL AND METHODS

2.1 Preoperative and intraoperative procedures

For preoperative and intraoperative procedures, task-based data were acquired. Resting-state networks were only evaluated intraoperatively. For task-based data, motor cortex stimulation was performed by repetitive, alternating left hand opening and closing at ≈ 1 Hz (successive periods of 20 s of rest followed by 20 s of stimulation). For resting-state data, patients stayed at rest.

Task-based MRI data were acquired before neurosurgery. A FLAIR sequence was used to obtain a 3D anatomical volume of the patient brain and an EG-EPI sequence combined with Statistical Parametric Mapping analysis (SPM)[11] was used to get functional brain maps.

Task-based and resting-state optical data were acquired during neurosurgery using RGB imaging. The videos were acquired with the wide field optical device described in Ref[7] after the patient's craniotomy and before the brain tumor resection surgery. ΔC_{HbO_2} and ΔC_{Hb} values were obtained with the modified Beer-Lambert law. Task-based maps were obtained using SPM analysis[11,12] and resting-state networks were investigated using seed correlation and independent component analysis (ICA).[13] The neurosurgeon also performed EBS after RGB imaging using a bipolar electrode (Nimbus Medtronic neurostimulator).

fMRI and optical task-based data were obtained with SPM analysis.[11,12] Thus, t statistics matrices were computed and binary matrices that indicated the extent of the functional brain area were obtained with the random field theory.[11]

Optical resting-state seed maps were obtained by comparing hemodynamic time curves measured at the patient's brain surface with the one measured at the level of a seed region using the Pearson correlation coefficient. The seed was represented as a 20 pixels diameter disk (1.2 mm diameter) located on the motor area identified by EBS. Optical resting-state ICA maps were computed with the FastICA algorithm. The objective was to estimate the spatial distribution of the hemodynamic sources (5 sources) during patient's rest. This was obtained by calculating the ICA mixing matrices. ICA matrices were sorted by selecting the one that best match fMRI and EBS identifications. Resting-state seed and ICA maps were thesholded to obtain a binary image that indicated the extent of the functional brain area. The thresholding operation only keeps pixels those values were above the theshold $T = \mu + \alpha.\sigma$. μ and σ are the mean and the standard deviation of the resting-state map, respectively. α is the severity criterion of the thresholding operation equal to 1.3.

2.2 Multi-modality registration procedure

A multi-modality registration procedure was set up to compare functional optical and MRI brain maps. These maps were co-registered to the preoperative anatomical MRI volume. First an affine transformation was computed to register the fMRI volume on the anatomical MRI volume using SPM.[11] Finally the functional optical map was registered on the anatomical MRI volume using a rigid transformation obtained by matching optical and MRI landmarks (least square minimisation).

2.3 Comparison between fMRI and functional optical maps

The DICE coefficient was computed between binary functional MRI and optical maps to evaluate the correspondence between the preoperative and intraoperative functional identifications. Resting-state procedure is likely to reveal the whole motor cortex, whereas a single body segment (e.g., face, hand) would be obtained with task-based procedure. To evaluate this case, the overlap coefficient[14] was computed between binary fMRI task-based maps (X in Eq. (1)) and binary optical maps (Y in Eq. (1)).

$$Overlap(X,Y) = \frac{|X \cap Y|}{|X|}, \tag{1}$$

where $|X|$ and $|Y|$ are the cardinalities of the two sets calculated after the thresholding operation.

3. RESULTS AND DISCUSSION

Intraoperative and preoperative identification of the patient's brain motor area using RGB imaging (task-based and resting state maps) fMRI (task-based maps) and EBS are represented in Fig. 1. The DICE and overlap coefficients computed between the task-based fMRI map and task-based, resting-state seed and ICA optical maps are indicated in table 1 and 2, respectively. The task-based optical map showed a good correspondence with the fMRI task-based map (DICE=0.75, Overlap=0.67). The optical resting-state seed map did not completely match the fMRI task-based map (DICE=0.58) due to greater functional identification with the resting-state seed map. However, resting-state procedure is likely to reveal larger functional areas than task-bask procedure. And, fMRI task-based map is almost totally included in the optical resting-state seed map (Overlap=0.76). The greater extent of the resting-state seed map could also be due to the threshold value, which will be more precisely defined in future works using SPM.[11,12] The resting-state ICA optical map showed a good correspondence with the fMRI task-based map (DICE=0.75, Overlap=0.81). The functional identifications obtained with preoperative fMRI and intraoperative optical imaging are consistent with those provided by EBS.

DICE(tb fMRI,tb Optics)	DICE(tb fMRI,rs seed Optics)	DICE(tb fMRI,rs ICA Optics)
0.75	0.58	0.75

Table 1. DICE coefficients computed between the task-based fMRI map (tb fMRI) and task-based (tb Optics) resting-state seed (rs seed Optics) and resting-state ICA (rs ICA Optics) optical maps represented in Fig. 1 A, C and D, respectively.

Overlap(tb fMRI,tb Optics)	Overlap(tb fMRI,rs seed Optics)	Overlap(tb fMRI,rs ICA Optics)
0.67	0.76	0.81

Table 2. Overlap coefficients (see Eq. (1)) computed between the task-based fMRI map (tb fMRI) and task-based (tb Optics) resting-state seed (rs seed Optics) and resting-state ICA (rs ICA Optics) optical maps represented in Fig. 1 A, C and D, respectively.

4. CONCLUSION

In this work, we presented a study comparing functional brain identification of a patient's motor area using preoperative task-based fMRI, intraoperative task-based and resting-state RGB imaging and EBS. Both task-based and resting-state RGB imaging showed a good correspondence with task-based fMRI and EBS identifications. These results indicated that RGB imaging could be used to complement EBS during neurosurgery and to improve the patient care process before, during and after neurosurgery. In future works, resting-state networks obtained with fMRI and optics will be compared. The multi-modality registration procedure could also be used to improve the detection of resting-state networks using RGB imaging by comparing resting-state networks obtained by fMRI.

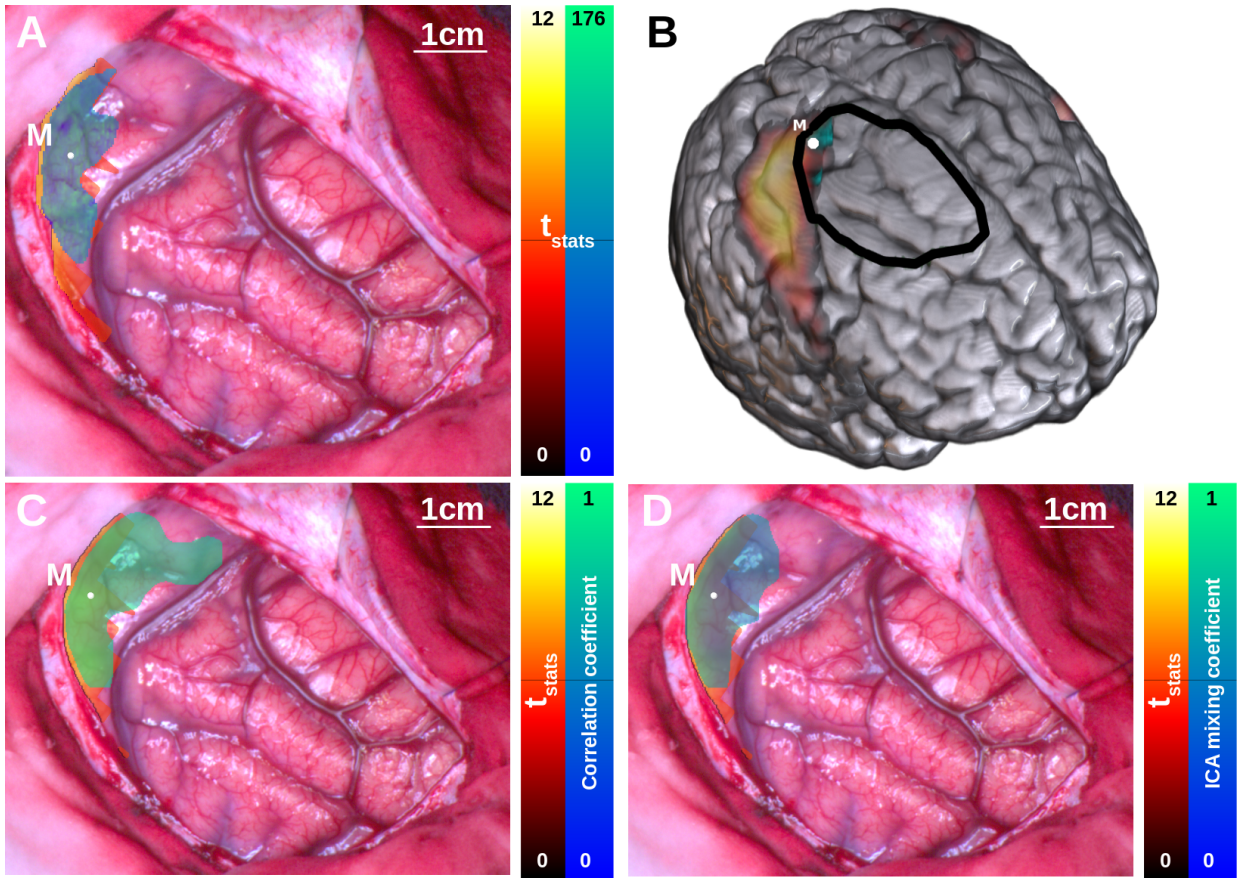

Figure 1. Intraoperative and preoperative identification of the patient's brain motor area using RGB imaging (blue to green maps) fMRI (red to yellow maps) and EBS (white point indicated with the letter M). A and B - Task-based RGB (ΔC_{Hb}) and fMRI map (A - in the RGB surgical window, B - in the fMRI volume). In image B, the black contour denotes the extent of the surgical window in image A. C - Resting state seed map using RGB imaging (ΔC_{Hb}). D - Resting state ICA map using RGB imaging (ΔC_{Hb}). In images A, C and D, the task-based fMRI map is also plotted.

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